

HAT-P-36b

R-0692

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260404 · created 2026-07-27T20:47:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461134.74 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.36 +/- 1.37 [%]
Orbital Inclination (inc)	80.21 +/- 2.82
Airmass coefficient 1 (a1)	1.025 +/- 0.058
Airmass coefficient 2 (a2)	-0.0 +/- 0.18
Scatter in the residuals of the lightcurve fit is	5.74 %
Best Comparison Star	#1 - [408, 78]
Optimal Aperture	2.82
Optimal Annulus	10.87
Transit Duration (day)	0.072 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.045 vs 0.126 (deviation 0.62σ)
Duration fit vs published	0.072 d vs 0.0925 d (deviation 0.98σ)
Mid-time O–C	15.5 min
Depth SNR	3.4
Residual scatter	5.74 %

Light curve

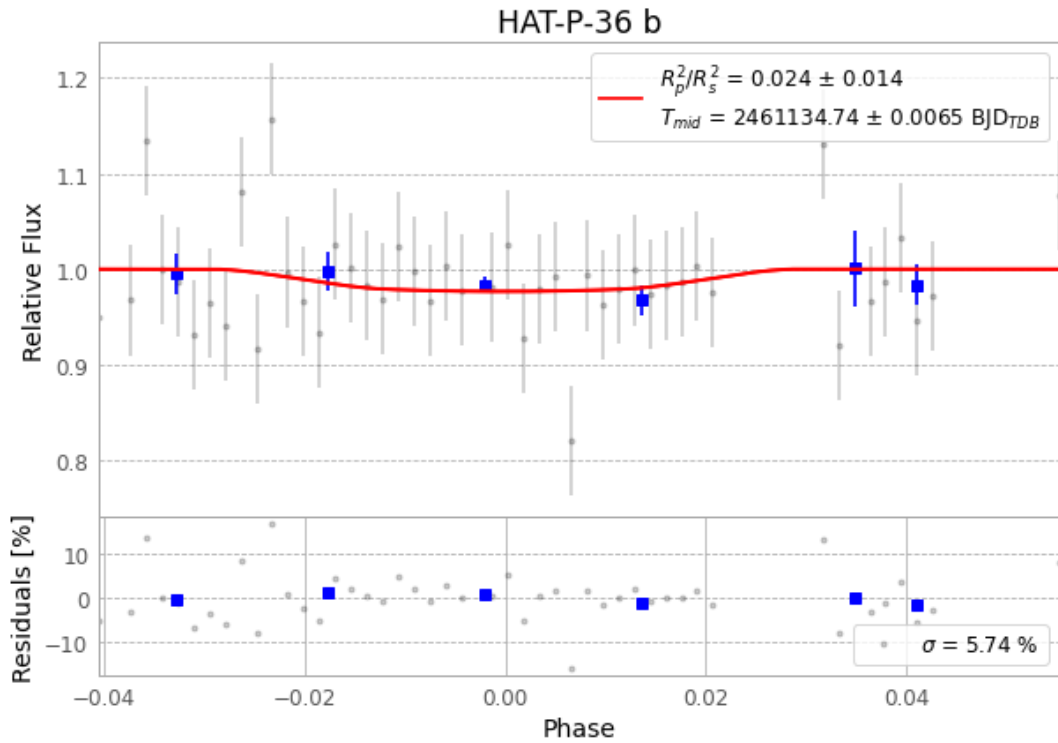


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-04-03.png (SHA-256 e98300e94e15be34...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 89230834311694a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

63 FITS frames (41 MB), HATP-36260404042115.FITS → HATP-36260404073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260404.log (f20f04dd3740...), FinalLightCurve_HAT-P-36 b_2026-04-03.png (e98300e94e15...), FinalLightCurve_HAT-P-36 b_2026-04-03.csv (c5183f669ffa...)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:47Z.